Experiment 8: Conduct Z test for the Data Given using Python Objective:

Aim:

To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. Null Hypothesis (H0): The average weight of the birds is 150 grams.
2. Alternative Hypothesis (H1): The average weight of the birds is not 150 grams.
3. Sample: Measure the weights of 30 birds randomly selected from the population.
4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.
5. Decision Rule: Use a significance level of α = 0.05.

Code:

```
import numpy as np
import scipy.stats as stats

# Define the sample data (hypothetical weights in grams)
sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152, 151,
148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])
# Population mean under the null hypothesis
population_mean = 150
# Calculate sample statistics
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation

# Number of observations
n = len(sample_data)
```

```
# Calculate the Z-statistic
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

# Calculate the p-value

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test

# Print results
print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")
# Decision based on the significance level
alpha = 0.05
if p_value < alpha:
print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")
else:
print("Fail to reject the null hypothesis: There is no significant difference in average weight from
150 grams.")
```

Output:

```
Sample Mean: 150.20
Z-Statistic: 0.6406
P-Value: 0.5218
Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```

Result:

The above code is executed successfully.